Vanguard ISG Model Portfolio Solutions Case Study

Ram Prasad

March 19, 2026

1 The Case for Global Diversification

The challenge is to dissuade performance chasing that naturally follows ex-post U.S. growth stock outperformance, and to remind the committee that global diversification remains a robust solution to uncertainty. I outline a communications framework predicated on four key ideas. First, distinguish ex-post realized returns from ex-ante expected returns. Second, argue, as done in the article[1], that performance chasing has no limiting principle. Third, frame diversification as insurance against concentration risk. Fourth, anchor non-U.S. allocations to a repeatable portfolio construction process.

Constructing a portfolio at time t uses information available up to and including time t, and thus by definition, realized returns over the coming decade are unknowable at time of construction. The relevant return is the conditional expectation $E_t(R_{t+1})$, not the realized return $R_{t-1,t}$. Return chasing is predicated on the belief that recent outperformance is sufficient evidence for future outperformance, i.e. $E_t(R_{t+1} \mid R_{t-1,t}) \approx R_{t-1,t}$. There is no theoretical foundation for the claim that recent outperformance is a sufficient statistic for future outperformance, and modest empirical evidence of return predictability agnostic of valuations[2] Even without formal econometrics, the logic of return extrapolation is self-refuting. It cannot simultaneously be the case that buying what recently outperformed guarantees strong future performance and that an asset outperformed, because if that asset was a laggard, then it cannot have outperformed the contemporaneous leading asset.

The argument for performance chasing has no limiting principle. The article notes that \$100 invested in U.S. stocks over the 10 years ended December 31, 2024, would have grown to \$334, versus \$160 for non-U.S. stocks, U.S. growth grew to \$470, the Magnificent Seven outperformed the information technology sector by 6.8 times, and Nvidia outperformed the Magnificent Seven by 6.3 times. Thus the logical conclusion of tilt towards more of what just won is a fully concentrated portfolio in the single best-performing stock, which is clearly untenable. Past outperformance cannot be the determinant, because by that logic diversification is indefensible at any level—country, sector, or security. If the committee would reject a 100% single stock portfolio, it has accepted diversification has value independent of recent realized returns, and thus the question is how much to diversify, not whether diversification itself is invalid.

[1] Vanguard, *Think differently about global diversification*, January 22, 2025.

[2] See Vanguard's discussion of valuation and expected returns in *Making the case for international equity allocations*, May 5, 2023, and *Building resilient portfolios through diversification*, March 26, 2024. These notes argue that the drivers of U.S. outperformance over the last decade have likely lowered forward-looking relative returns.

Diversification can be thought of as a form of insurance using expected-utility. If an investor can choose between a risky payoff of 5% or 13% with equal probability and a certain 9% outcome, most investors prefer the 9%, even though both have the same expected value. Investors are typically risk-averse with a strictly concave utility function $u(\cdot)$, and by Jensen's inequality, $u(E[X]) \geq E[u(X)]$. If X is the risky payoff and $CE(X)$ solves $u(CE(X)) = E[u(X)]$, then for a risk-averse investor $CE(X) < E[X]$ and a mean-preserving reduction in risk has value. Global diversification is a mechanism for reducing concentration risk, and non-U.S. exposure is portfolio insurance against the possibility that U.S. concentration performance proves to be cyclical rather than durable.[3]. Portfolio construction concerns downside, path dependence, and survivability, not merely maximizing expected returns. Catastrophic losses, once realized, cannot be diversified away after the fact.

Non-U.S. allocation follows from a rigorous portfolio construction process. Current market conditions make diversification outside the U.S. more attractive[4]. Maintaining or increasing non-U.S. weights today is fundamentally a valuation-aware rebalancing toward regions that have cheaper starting points and lower concentration risk.[5] The communication framework thus begins with acknowledging regret over missed performance, reframing the benchmark for performance as not the best realized return but outcomes consistent with the client's long-run objective and risk tolerance, reiterating the role of each sleeve (e.g. U.S. equities provide exposure to the world's largest equity market, developed ex-U.S. equities diversify valuation, sector, and policy regimes, emerging markets provide access to different growth and demographic exposures), using forward-looking language and arguing the past decade in and of itself doesn't determine the next decade's returns, showing that concentration is a risk-budgeting decision (specifically, reduction in non-U.S. exposure increases exposure to U.S. macro, valuation, policy, and sector concentration risks), and committing to allocation bands and disciplined (e.g. rule-based, period rebalancing) reviews.

[3]Vanguard makes the point that diversification acknowledges that we cannot predict the future and hedges that uncertainty. See Vanguard, *Diversify the way you think about diversification*, September 11, 2024.

[4]Vanguard argues that the same drivers of U.S. outperformance over the last decade, e.g. richer valuations and greater concentration in mega-cap growth, may have lowered forward-looking relative returns for U.S. equities and improved the case for international equities. See Vanguard, *Making the case for international equity allocations*, May 5, 2023. See also Vanguard, *Building resilient portfolios through diversification*, March 26, 2024, which argues for stronger forward-looking international equity return prospects based on cheaper valuations.

[5]See also Vanguard, *Think differently about global diversification*, January 22, 2025, and *Less-concentrated markets could buoy active managers*, April 17, 2025, for discussion of the risks created by concentration in mega-cap growth.

2 Currency Hedging for Global Bonds

Global bonds can outperform U.S. bonds for several reasons, including a foreign premium, geographic diversification, market breadth, and currency hedging[6]. Global bonds bring exposure to adverse and possibly favorable movements in foreign exchange rates. A natural question is how much of that exposure should a U.S.-based investor hedge, and under what circumstances should they deviate. In this section, I recommend a **100% hedge ratio** because full hedging preserves the intended role of fixed income in the portfolio, minimizes FX-driven volatility, and captures hedging carry arising from interest rate differentials. Accordingly, deviations should generally reflect deliberate macro views or unusual carry conditions. To motivate my recommendation, I derive a mathematical framework for USD returns on foreign bonds with partial hedging. A full derivation can be found in Appendix B, I state key results below.

2.1 USD Returns on Foreign Bonds

Let $R^F_{t,t+1}$ denote the foreign currency bond return and $R^{FX}_{t,t+1}$ the FX return. The gross USD return on an unhedged foreign bond position can be written as

$$1 + R^{USD}_{t,t+1} = (1 + R^F_{t,t+1})(1 + R^{FX}_{t,t+1}).$$

Note the following. First, USD returns are multiplicative in the foreign bond return and the FX return. If the bond appreciates in foreign currency terms and the foreign currency appreciates relative to the USD, the FX return amplifies the USD return. If the foreign currency depreciates, the FX return dampens the USD return. Second, the random variables driving returns are the future bond price P^F_{t+1} and the future spot exchange rate S_{t+1}. Thus the foreign bond portfolio has exposure to two sources of risk, bond risk (interest rate risk, credit risk) and currency risk.

Currency hedging removes uncertainty associated with the future spot rate. The standard hedging instrument is an FX forward contract, in which two parties agree to transact at a future date at a pre-specified exchange rate. Let h_t denote the hedge ratio. With partial hedging, the gross USD return becomes

$$1 + R^{USD}_{t,t+1} = (1 + R^F_{t,t+1}) \left[h_t \frac{F_{t,t+1}}{S_t} + (1 - h_t)(1 + R^{FX}_{t,t+1}) \right],$$

where $h_t = 0$ corresponds to a fully unhedged position and $h_t = 1$ corresponds to a fully hedged position.

[6] See Vanguard, *Global bonds: Four key drivers of outperformance*, October 8, 2024.

2.2 Hedging Carry

The forward exchange rate reflects interest rate differentials via covered interest parity (CIP),

$$\frac{F_{t,t+1}}{S_t} = \frac{1 + i_t^{USD}}{1 + i_t^F}.$$

This relationship holds as a result of no-arbitrage and introduces a carry component. If USD interest rates exceed foreign rates, the forward rate exceeds the spot rate and hedging increases the USD return relative to the foreign currency return. As noted in the article, this explains why low-yield foreign bonds can produce attractive USD hedged returns. For example, a one-year Japanese government bond yielding 0.2% in JPY corresponds to roughly 4.7% when hedged into USD.

2.3 Basis Risk

Covered interest parity generally holds but can deviate due to funding pressures. Let b_t denote the cross-currency basis:

$$\frac{F_{t,t+1}}{S_t} = \frac{1 + i_t^{USD}}{1 + i_t^F}(1 + b_t).$$

When $b_t \neq 0$, basis risk affects returns. Even if the investor is fully hedged to a single-period's exposure, multi-period portfolios require rolling over forward contracts and thus realized returns depend on future forward rates. In practice, basis risk is much smaller than FX risk. FX volatility is on the order of 10% annualized, compared with roughly 3–5% volatility for investment-grade bonds, and cross-currency basis volatility is usually measured in tens of basis points.

2.4 Portfolio Volatility

The appendix derives a complete expression for the conditional variance of USD returns as a function of the hedge ratio. Using a first-order approximation, USD return variance decomposes as follows:

$$\mathrm{Var}(R^{USD}) \approx \text{Bond variance} + (1-h)^2 \text{FX variance} + \text{covariance terms}.$$

Note that FX risk scales with $(1-h)^2$, thus increasing the hedge ratio reduces the FX contribution to total portfolio volatility.

2.5 Recommended Hedge Ratio

With the key mathematical results established, I state the case for a U.S.-based investor holding global aggregate bonds maintaining a **100% hedge ratio**.

First, the primary reason to hold global bonds is to obtain bond-like returns commensurate with bond-like volatility, while benefiting from diversification across countries. Leaving any FX exposure unhedged introduces additional volatility unrelated to bond fundamentals. Full hedging lowers

portfolio volatility while avoiding drift in the investment objective. Moreover, currency movements often correlate with global macro shocks. For example, during risk-off periods, investors move toward U.S. Treasuries, strengthening the USD and weakening the foreign currency (relative to the USD), and thus foreign bond returns in USD terms can fall even if local currency bond returns remain stable. Hedging eliminates this procyclical FX amplification and preserves the defensive role of bonds in the portfolio.

Second, complete hedging maximizes the carry benefit. Hedging embeds interest rate differentials through the forward rate, and when U.S. interest rates exceed foreign rates, hedging increases the USD return relative to the foreign currency return. A partial hedge captures part of this carry benefit while still leaving significant FX risk in the portfolio. The exact tradeoff is an empirical question, but it is indisputable that a partial hedge neither fully reduces FX volatility nor passes through the full carry benefit, and thus is a compromise.

Third, currency hedging determines whether the four levers of excess global bond returns get cleanly transmitted to USD investors. Full hedging removes the multiplicative FX term. When $h_t = 1$ the USD return depends only on the foreign bond return and the hedging carry, which allows the investor to capture the benefits of geographic diversification, foreign premia, and broader market opportunities. Empirically, MSCI index analysis shows that U.S.-dollar investors who hedged their currency exposure over the past 14 years experienced lower risk and higher returns compared with unhedged investors[7] and Vanguard notes currency hedging isolates credit and duration exposures[8].

2.6 The Case for Deviations

The primary downside of hedging is basis risk. Hedging is not costless, and deviations from CIP can make hedging more expensive than implied by interest rate differentials. As stated earlier, basis risk is comparatively small, relative to FX volatility, thus I argue it is worth bearing. Nonetheless, there are cases in which deviating from a 100% hedge ratio is justifiable. First, if the investor holds a deliberate macro view on FX and wishes to exercise it, they may deviate. For example, if the investor believes the USD is significantly overvalued or will otherwise depreciate (e.g. as a result of policy instability, loss of confidence in the central bank and unanchored inflation expectations), leaving some currency exposure unhedged may provide additional returns. The investor not only needs confidence in their macro bet, but confidence that they will be compensated fairly for bearing the FX risk. Second, if forward carry is unusually unattractive, such as when foreign interest rates exceed U.S. rates or when cross-currency basis conditions make USD hedging unusually expensive, they may deviate. Finally, the investor may deviate if they expect basis risk to be unusually volatile over the lifetime of the hedge. In each case, the deviation is tactical.

[7] MSCI, *Markets in Focus: Currency Risk Higher for Global Equity Investors*, July 9, 2024.

[8] Vanguard Investment Strategy Group, *International fixed income: Currency hedging considerations*, 2022.

3 Time-Varying Asset Allocation

Vanguard's VCMM provides 10 and 30-year horizon forecast distributions for expected returns and volatilities across a range of asset and sub-asset classes[9]. I use these results and follow Vanguard's approach for time-varying asset allocation[10] to evaluate the case for tilts in allocations to Developed ex-U.S. and EM Equities. I proceed as follows. First, I identify an equity policy portfolio as a benchmark for suballocations. Second, I compare medium-horizon VCMM forecasts with long-run forecasts and identify signals for a time-varying tilt. Third, I infer investor risk aversion from the policy portfolio. Fourth, I solve a mean-variance optimization problem using VCMM forecasts and inferred risk aversion from step 3. Fifth, I stress test my cross-asset correlation assumptions. Sixth, I evaluate tail risk using VCMM quantile diagnostics. Seventh, I outline guardrails to preserve policy discipline. Finally, I define a review cadence.

3.1 Policy Portfolio

A TVAA makes bounded deviations from a strategic asset allocation, thus I begin by identifying an SAA. For global equities, the cleanest benchmark is global market capitalization, because that is the aggregate portfolio the market already holds. Figure 2 shows the regional composition of Vanguard Total World Stock Index Fund[11] which holds 64.4% in North America and 10.3% in emerging markets, with the residual allocated to non-U.S. developed regions. For simplicity, and treating the suballocation to equities as the total portfolio, I assume the policy equity allocation follows global market capitalization and round the benchmark to 60% U.S. equities, 30% developed ex-U.S. equities, and 10% emerging markets equities. This portfolio represents a long-run equilibrium, or steady-state, allocation, and is the basis for inference about investor risk aversion.

Table 1: Global Equities Policy Portfolio

Asset	Policy Weight
U.S. equities	60%
Developed ex-U.S. equities	30%
Emerging markets equities	10%

3.2 VCMM Return Distributions

I examine the VCMM return distributions at the 10-year horizon and the 30-year horizon, taking the 30-year horizon as the steady-state return environment, and treating deviations from that steady state (e.g. cases in which the 10-year forecast differs from the 30-year forecast) as a candidate

[9] See Vanguard ISG, *Vanguard Capital Markets Model (VCMM) return forecasts*, accessed 2026.

[10] Todd Schlanger, Roger A. Aliaga-Díaz, Cheng Li, and Harshdeep Ahluwalia, *Time-Varying Asset Allocation: Vanguard's Approach to Dynamic Portfolios*, The Vanguard Group, October 2024.

[11] Vanguard Total World Stock Index Fund portfolio composition: `https://investor.vanguard.com/investment-products/mutual-funds/profile/vtwax#portfolio-composition`, accessed 2026.

signal for a time-varying tilt[12]. The premise is that over a 10-year horizon, asset classes can deviate materially from long-run fair value, and that fair value is tied to future realized returns. An asset with a cheaper (more expensive) starting valuation should have higher (lower) future realized returns, as total return depends in part on the starting yield and valuation multiple, so cheaper (more expensive) assets have more room for multiples to mean revert upward (downward).

Table 2 shows VCMM expected return forecasts at the 10-year and 30-year horizons. U.S. equities have median returns of 4.9% and 5.6%, developed ex-U.S. equities have median returns of 5.9% and 7.0%, and emerging markets equities have median returns of 4.2% and 6.1%, respectively.

Table 2: VCMM Median Equity Return Forecasts, 10-year and 30-year Horizons

Asset	10Y	30Y
U.S. equities	4.9%	5.6%
Developed ex-U.S. equities	5.9%	7.0%
Emerging markets equities	4.2%	6.1%

Note the following. First, U.S. equities are expected to return less over the 10-year horizon than over the 30-year horizon. I interpret that as evidence that U.S. equities are expensive today relative to their long-run equilibrium, as the 10-year forecast reflects returns conditional on current valuations, whereas the 30-year forecast is closer to a long-run fair-value return. Second, the spread between developed ex-U.S. equities and U.S. equities is a key valuation signal. At the 10-year horizon, the spread is $5.9\% - 4.9\% = 1.0\%$, while at the 30-year horizon, the spread is $7.0\% - 5.6\% = 1.4\%$. Developed ex-U.S. equities are cheap relative to U.S. equities, and the long-run return premium of developed ex-U.S. over U.S. equities is actually somewhat larger in steady state than at the medium horizon.

The positive 10-year spread indicates developed ex-U.S. equities are priced to deliver about 1 percentage point more in annual return than U.S. equities over the coming decade, which compounded over 10 years implies roughly 10% cumulative outperformance:

$$\left(\frac{1.059}{1.049}\right)^{10} - 1 \approx 9.95\%.$$

Over a 30-year horizon, the 1.4 percentage point spread translates to much larger cumulative outperformance:

$$\left(\frac{1.070}{1.056}\right)^{30} - 1 \approx 48.5\%.$$

Even an ostensibly modest annual spread compounds into meaningful performance differences. I note a somewhat subtle economic interpretation of the spread's magnitudes. The 10-year developed

[12] Treating the 30Y median expected return as equilbrium or fair value and the 10Y median as a valuation affected return is an assumption I make for the purposes of the case. The difference proxies medium-term value dispersion, in lieu of a formal decomposition into equilbrium and valuation components.

ex-U.S. versus U.S. spread is positive, suggesting a tilt toward developed ex-U.S. equities, but the 10-year spread is *smaller* than the 30-year spread. Thus the medium-horizon valuation signal is positive, but not stronger than in steady state. Developed ex-U.S. equities appear to have a structural return premium relative to U.S. equities, but current valuation conditions do not widen that premium beyond its long-run level. I conclude the case for overweighting developed ex-U.S. equities is modest, and doesn't warrant an aggressive deviation from policy.

3.3 Investor Risk Aversion

Mean-variance optimization requires a risk aversion parameter. In lieu of using a value from academic or practitioner publications, I infer the parameter, γ, from the observable policy portfolio, VCMM expected returns, and volatilities. Formally, the idea is to choose γ such that the policy portfolio solves

$$w^{\text{policy}} = \arg\max_{w} \left\{ \mu'_{30Y} w - \frac{\gamma}{2} w' \Sigma w \right\},$$

where

$$\mu_{30} = (5.6,\ 7.0,\ 6.1),$$

corresponds to U.S. equities, developed ex-U.S. equities, and emerging markets equities, and policy weights

$$w^{\text{policy}} = (0.60,\ 0.30,\ 0.10).$$

When input into the model, the calibrated parameter reproduces the policy portfolio evaluated using steady-state (30-year VCMM) forecasts. Put differently, the observed policy allocation coupled with VCMM forecasts reveals investor risk aversion. I use calibrated $\hat{\gamma}$ in subsequent optimization. Absent a correlation matrix, calibration is difficult. I thus make plausible steady-state covariance assumptions to construct the covariance matrix, starting with the 30-year volatilities

$$(17.0,\ 20.1,\ 27.9)$$

and a plausible intra-equity correlation matrix

$$\begin{pmatrix} 1 & 0.85 & 0.75 \\ 0.85 & 1 & 0.80 \\ 0.75 & 0.80 & 1 \end{pmatrix}.$$

Under these assumptions, the implied risk-aversion parameter is $\gamma \approx 0.19$ (See Appendix D for the calibration procedure). Stress testing my assumptions with a correlation grid gives a narrow band of $\gamma \in [0.17,\ 0.21]$. This is a moderate level of risk aversion, and indicates the TVAA should be sensitive to expected return differences but not overly so. A very low γ may justify aggressive tilts, while a higher γ may warrant a TVAA much closer to policy.

3.4 Mean-Variance Optimization using VCMM Forecasts

Using the policy portfolio and the calibrated risk-aversion parameter, I solve the mean-variance problem using VCMM return forecasts. Computing the optimal allocation indicates whether the forecast return differentials are strong enough to justify a tilt away from the policy portfolio, and if so, in which direction. Before solving, I compare the 10-year risk-adjusted returns of the three equity blocks using the 10-year VCMM median cash rate of 3.5% as a proxy for the risk-free rate. The "Sharpe" ratios are approximately 0.092 for U.S. equities, 0.132 for developed ex-U.S. equities, and 0.027 for emerging markets equities.

Table 3: 10-year VCMM Equity Risk-Adjusted Returns

Asset	Sharpe-like ratio (10Y)	Volatility
U.S. equities	0.092	15.2
Developed ex-U.S. equities	0.132	18.2
Emerging markets equities	0.027	25.6

Note the following. First, developed ex-U.S. equities have the highest risk-adjusted return, while emerging markets have a very poor Sharpe ratio. Even without assumed correlations, I would expect the optimizer to overweight developed ex-U.S. equities, underweight emerging markets, and modestly underweight U.S. equities. Second, and moreover, the comparison between developed ex-U.S. and emerging markets is robust to assumed correlation. Developed ex-U.S. weakly dominates emerging markets on a simple return-versus-volatility basis at both the 10-year and 30-year horizons. At 10 years, developed ex-U.S. has a higher median return (5.9% versus 4.2%) at a lower median volatility (18.2% versus 25.6%). At 30 years, developed ex-U.S. again has a higher median return (7.0% versus 6.1%) at lower median volatility (20.1% versus 27.9%). Third, and consequently, there is no standalone case for an emerging markets overweight. An optimizer may prescribe a meaningful allocation to emerging markets if there is a strong diversification benefit from unusually low or negative correlation with the other equity blocks, but in practice that is simply not the case. Intra-equity market correlations are not low enough to overcome EM's first-order disadvantage in both return-volatility and Sharpe ratio. Fourth, the comparison between U.S. and developed ex-U.S. equities is different. Developed ex-U.S. has both a higher expected return and a higher volatility than U.S. equities, so the allocation between the two is not correlation-invariant. In particular, when correlations are high, an optimizer naturally tends to push more weight toward the higher return asset, namely developed ex-U.S. equities.

I solve the long-only mean-variance optimization problem

$$\max_{w} \; \mu' w - \frac{\gamma}{2} w' \Sigma w$$

subject to

$$\mathbf{1}' w = 1, \qquad w_i \in [0, 1].$$

As a first pass and robustness check, I solve the problem without guardrails, using the assumed covariance matrix from the calibration step. Under the 30-year expected return vector, the optimizer produces a corner solution of 100% to developed market ex-U.S. equities, exactly as expected and confirming the basic directional signal. However, this is not a realistic TVAA recommendation. A time-varying asset allocation is supposed to make bounded deviations around the policy portfolio, not replace the strategic allocation with a new one. To preserve policy discipline, I therefore impose explicit guardrails around the benchmark weights:

$$w_{\text{US}} \in [0.55, 0.65], \qquad w_{\text{DM}} \in [0.25, 0.35], \qquad w_{\text{EM}} \in [0.07, 0.13],$$

corresponding to ±5 percentage points around the policy weights for U.S. and developed ex-U.S. equities, and ±3 percentage points for emerging markets equities (by design, bands are proportional to the weights)[13].

I solve the now constrained optimization problem using both the 30-year and 10-year expected return vectors. Table 4 reports the 10-year and 30-year allocations, along with portfolio moments.

Table 4: Mean–Variance Optimization with Guardrails

Asset / Statistic	30Y Horizon	10Y Horizon
U.S. equities	55%	58%
Developed ex-U.S. equities	35%	35%
Emerging markets equities	10%	7%
Expected return	6.14%	6.12%
Volatility	18.08%	17.87%

Note the following. First, using VCMM 30-year expected returns, the TVAA pushes developed ex-U.S. equities to the upper bound, increasing the allocation from 30% to 35%, and cuts U.S. equities from 60% to 55%. Emerging markets remain at the policy weight of 10%. This result is entirely consistent with a structural long-run developed ex-U.S. return premium over U.S. equities. The optimizer increases developed ex-U.S. exposure as much as the policy guardrails allow, but it does not change the EM allocation. Second, using VCMM 10-year expected returns, the TVAA again pushes developed ex-U.S. equities to the upper bound, but now it reaches that overweight partly by reducing both U.S. and EM allocations. Relative to policy, the allocation to developed ex-U.S. rises by 5 percentage points, U.S. equities fall by 2 percentage points, and emerging markets fall by 3 percentage points to their lower bound. Third, both 10-year and 30-year constrained solutions push developed ex-U.S. equities to the top of the guardrail range, indicating that case for a positive tilt to developed ex-U.S. is robust. Finally, the medium-horizon valuation signal does *not*

[13] Moreover, guardrails constrain deviations from policy and hedge against estimation error and model risk inherent to VCMM forecasts. Additional constraints include tracking error, e.g. TE $\leq$ 3%, and volatility bounds, e.g. $\sigma_p \leq \sigma_{\text{policy}} + 0.02$. Constraints prevent the TVAA from overreacting to noisy forecast inputs and keep the TVAA's risk profile close to that of the benchmark.

justify a broad overweight to all non-U.S. equities. Rather, it justifies a *selective* tilt, overweight developed ex-U.S. equities and funds that overweight by trimming both U.S. and EM exposures, with the bulk of that funding from EM because of its very poor medium-horizon risk-adjusted outlook.

I conclude that the case for a developed ex-U.S. overweight relative to the policy benchmark is robust across correlation and model specifications but modest in magnitude. There is no case for an EM overweight. In fact, the 10-year VCMM forecasts support an EM underweight within guardrails, in the absence of which the optimizer collapses to a corner solution representing a new SAA, not a TVAA. Thus I allow a developed ex-U.S. tilt when the 10-year forecast return advantage over U.S. equities is positive and economically meaningful, and I allow an EM underweight when EM has both a poor standalone Sharpe ratio and is weakly dominated by developed ex-U.S. in return-volatility space. Both conditions are satisfied, hence the prescribed tilt.

3.5 Stress Testing Assumed Correlations

The previous optimization relies on a single assumed correlation matrix. Because correlation assumptions can be an unreliable input in portfolio construction, I stress test the results using a grid of plausible correlation values. Specifically, I vary the three pairwise correlations across reasonable ranges for global equity markets:

$$\rho_{\text{US,DM}} \in [0.80, 0.90], \qquad \rho_{\text{US,EM}} \in [0.65, 0.80], \qquad \rho_{\text{DM,EM}} \in [0.70, 0.85].$$

The ranges reflect the empirical fact that correlations across major equity markets are typically high and tend to increase during global risk episodes, and I construct the grid to capture moderate variation around a realistic baseline rather than extreme scenarios. For each correlation triple, I construct the covariance matrix $\Sigma = D\rho D$, recompute the implied risk-aversion parameter γ using the calibration procedure from the previous section, and then resolve both the unconstrained and guardrailed mean-variance optimization problems.

Table 5 reports results of the correlation grid exercise. Across all 48 correlation configurations, developed markets ex-U.S. gets 100% allocation. Using guardrails, the optimization produces very stable allocations across the entire grid. Developed ex-U.S. equities always sit at the upper bound of the allowed range, increasing from the policy weight of 30% to the maximum allowed 35%. The remaining adjustment occurs through modest shifts between U.S. and emerging markets equities.

Table 5: Guardrailed Portfolio Weights Across Correlation Grid, 30-year VCMM Forecasts

Statistic	U.S. Equities	DM ex-U.S. Equities	EM Equities
Count	48	48	48
Mean	0.5561	0.3500	0.0939
Std. Dev.	0.0107	0.0000	0.0107
Min	0.5500	0.3500	0.0712
25%	0.5500	0.3500	0.0945
50%	0.5500	0.3500	0.1000
75%	0.5555	0.3500	0.1000
Max	0.5788	0.3500	0.1000

U.S. equities range between roughly 55% and 57.9%, while emerging markets equities vary between about 7.1% and 10%. Across all correlation assumptions, the optimizer consistently pushes developed ex-U.S. equities to the top of the band, pushes U.S. equities toward the lower end of their band, and allows emerging markets to absorb the remaining adjustment[14].

3.6 Tail Risk

VCMM forecast tables do not report higher moments such as skewness or kurtosis directly. However, they do provide percentile forecasts, which I use to construct simple shape diagnostics for the return distribution. Specifically, I compute three quantile-based metrics:

$$\text{Downside} = P50 - P5, \qquad \text{Upside} = P95 - P50, \qquad \text{Skew} = \text{Upside} - \text{Downside}.$$

Downside measures how far the left tail sits below the central forecast (smaller is better), upside measures how far the right tail sits above the central forecast (larger is better), and skew measures whether the simulated distribution fans out more on the upside or downside. Table 6 shows that Developed ex-U.S. equities exhibit slightly better left-tail containment than U.S. equities and substantially better containment than emerging markets. Emerging markets display the largest downside dispersion, consistent with their much higher volatility.

[14]This behavior is economically intuitive. Developed ex-U.S. equities combine the highest expected return with a volatility level not dramatically higher than U.S. equities. As correlations rise, diversification benefits across equity subclasses weaken, which increases the optimizer's tendency to allocate toward the higher expected return asset. Emerging markets, by contrast, are penalized by their low expected return and high volatility, thus they remain in the portfolio on account of the policy guardrails.

Table 6: 10-year Return Distribution Asymmetry (VCMM)

Asset	Downside ($P50 - P5$)	Upside ($P95 - P50$)	Skew
U.S. equities	8.4	9.0	0.6
Developed ex-U.S. equities	7.4	8.0	0.6
Emerging markets equities	9.7	10.9	1.2

Table 7: 30-year Return Distribution Asymmetry (VCMM)

Asset	Downside ($P50 - P5$)	Upside ($P95 - P50$)	Skew
U.S. equities	5.8	6.0	0.2
Developed ex-U.S. equities	4.0	4.1	0.1
Emerging markets equities	6.2	6.7	0.5

EM equities have the largest upside tail, commensurate with their larger downside risk. Over a 30-year horizon, dispersion narrows relative to the 10-year horizon, but emerging markets continue to exhibit the largest upside asymmetry, reflecting greater long-run return uncertainty. For an investor with moderate risk-aversion, this pattern again supports a tilt toward developed ex-U.S. equities and away from EM and reinforces the Sharpe ratio comparison provided above.

3.7 Recommended TVAA

My recommendation reflects a confluence of factors. Using the 10-year expected returns, volatilities, and the assumed covariance matrix, the optimizer recommends approximately $w = (0.58, 0.35, 0.07)$, corresponding to shifts of $(-2, +5, -3)$ percentage points relative to policy. However, the magnitude of the shift is contingent on the strength of the valuation signal relative to the long-run equilibrium. The 10-year developed ex-U.S. versus U.S. return spread is 1%, but the 30-year spread is larger at 1.4%, which as stated earlier indicates developed ex-U.S. equities have a structural return premium and the policy portfolio already reflects this long-run expectation.

Consequently, I implement only about one-half to two-thirds of the optimizer's suggested shift. Half of the move corresponds to $(-1, +2.5, -1.5)$ percentage points relative to policy, which yields an allocation of approximately (58–59, 33–34, 8–10). Table 8 shows my final recommendation.

Table 8: Policy Portfolio and Recommended TVAA

Asset	Policy	Guardrail Band	TVAA Recommendation	Change vs Policy
U.S. equities	60%	55–65%	58–59%	−2 to −1 pp
Developed ex-U.S. equities	30%	25–35%	33–34%	+3 to +4 pp
Emerging markets equities	10%	7–13%	8–10%	−2 to 0 pp

This allocation, a moderate overweight to developed ex-U.S. equities, a small underweight to U.S. equities, and no overweight to emerging markets, remains within all guardrails, follows the optimizer's direction, and responds to the medium-horizon valuation signal without ignoring the steady-state anchor.

3.8 Conditions for an EM Tilt

The following conditions may (individually) merit a tilt towards emerging markets equities:

1. 10-year EM Sharpe Ratio $\geq$ 10-year U.S. Sharpe Ratio.
2. Expected return spread exceeds 2 percentage points:

$$E[R_{EM}] - E[R_{US}] \geq 2\%.$$

3. EM downside risk is close to, and not materially worse than, that of U.S. equities:

$$(P50 - P5)_{EM} \leq (P50 - P5)_{US} + 1\%.$$

These conditions correspond to superior risk adjusted returns, meaningfully superior expected returns[15], and comparable downside risk. Using current 10-year VCMM forecasts, EM equities fail all three tests. Relative to U.S. equities, EM shows a return shortfall of $4.2 - 4.9 = -0.7$, a volatility difference of $25.6 - 15.2 = 10.4$, and a downside risk spread of $9.7 - 8.4 = 1.3$.

3.9 Rebalancing and Review Cadence

Finally, the TVAA requires a disciplined review process. I review the VCMM forecasts quarterly and rerun the optimization at the same frequency. However, the portfolio is rebalanced annually to avoid unnecessary transaction costs while still keeping allocations reasonably close to target weights. Quarterly review strikes a reasonable balance as it avoids reacting to high-frequency noise while still responding to meaningful changes in the return opportunity set.

[15] For EM to match the U.S. Sharpe ratio, it would require roughly 1 percentage point more in expected returns, and given the weaker tail profile and higher implementation costs associated with EM markets, I require an additional percentage point as cushion.

4 Portfolio Construction

In this section, I construct a portfolio for a U.S.-based investor, with a moderate risk tolerance, 10-year investment horizon, a taxable account, and the asset universe comprised of U.S. equities, developed ex-U.S. equities, emerging markets equities, U.S. bonds, and global ex-U.S. bonds. I use the VCMM 10-year forecasts for expected returns and volatilities (See Appendix A) and start with a baseline portfolio of 60% equities, 40% bonds, with the equities sub-allocated as 80% to U.S. equities, 20% to non-U.S. equities, and 100% of the bond allocation to U.S. bonds. Based on my analysis in section 2, I assume a 100% hedge ratio for any non-U.S. bond exposure.

4.1 Baseline Portfolio Construction

Unspecified in the baseline portfolio is the composition of the non-U.S. equities allocation. The asset universe includes developed ex-U.S. and emerging markets, thus I first identify the composition. I begin with the investor's current policy portfolio $w^{\text{base}} = (0.60 \text{ equities}, \ 0.40 \text{ bonds})$. An equity sleeve allocated 80% to U.S. equities and 20% to non-U.S. equities, and a bond sleeve allocated entirely to U.S. aggregate bonds implies top-level portfolio weights are

$$w_{US,eq} = 0.60 \times 0.80 = 0.48, \qquad w_{Int,eq} = 0.60 \times 0.20 = 0.12, \qquad w_{US,bond} = 0.40.$$

For the split of the non-U.S. equity sleeve between developed markets ex-U.S. (DM) and emerging markets (EM), I consider three plausible and correlation agnostic approaches: market composition, equal weighting, and risk parity[16].

1. Market Composition

First, I use a market-capitalization split for the non-U.S. equity sleeve. As a practical proxy, I use the Vanguard Total International Stock ETF (VXUS), whose regional allocation reports emerging markets at 26.60% of the non-U.S. sleeve, implying developed ex-U.S. markets at 73.40%.[17] Thus

$$w_{DM|Int} = 0.734, \qquad w_{EM|Int} = 0.266,$$

with implied total portfolio weights

$$w_{DM} = 0.12 \times 0.734 = 0.08808, \qquad w_{EM} = 0.12 \times 0.266 = 0.03192.$$

[16] Optimization results in this section should be interpreted as illustrative of directional tilts. The results are contingent on simplifying assumptions and expected return, correlation, and risk tolerance inputs.

[17] Vanguard, VXUS fund page, accessed 2026.

2. Equal Weighting

Split the non-U.S. sleeve evenly between developed and emerging markets,

$$w_{DM|Int} = w_{EM|Int} = 0.50,$$

which yields

$$w_{DM} = w_{EM} = 0.12 \times 0.50 = 0.06.$$

3. Risk Parity

Risk parity allocates inversely proportional to volatility. Using the VCMM 10-year median volatilities, $\sigma_{DM} = 18.2\%, \sigma_{EM} = 25.6\%$, the inverse-volatility weights are

$$\tilde{w}_{DM} = \frac{1/\sigma_{DM}}{1/\sigma_{DM} + 1/\sigma_{EM}} \approx 0.5845, \qquad \tilde{w}_{EM} = \frac{1/\sigma_{EM}}{1/\sigma_{DM} + 1/\sigma_{EM}} \approx 0.4155.$$

Applying these weights to the international sleeve gives

$$w_{DM} = 0.12 \times 0.5845 \approx 0.0701, \qquad w_{EM} = 0.12 \times 0.4155 \approx 0.0499.$$

Table 9 shows the resulting candidate baseline portfolios.

Table 9: Candidate Baseline Portfolio Allocations

Portfolio	U.S. Equities	DM ex-U.S. Equities	EM Equities	U.S. Aggregate Bonds
Market Composition	48.0%	8.81%	3.19%	40.0%
Equal Weight	48.0%	6.00%	6.00%	40.0%
Risk Parity	48.0%	7.01%	4.99%	40.0%

To evaluate these candidate portfolios and settle on the exact baseline portfolio allocation, I need cross-asset correlations. Published VCMM forecasts do not provide these, so I repeat section 3's covariance matrix construction exercise.

4.2 Correlation Grid and Covariance Construction

I evaluate a finite grid of plausible correlation matrices spanning empirically reasonable ranges. For the three equity regions, I consider the candidate values

$$\rho_{US,DM} \in \{0.75, 0.85, 0.95\}, \quad \rho_{US,EM} \in \{0.65, 0.75, 0.85\}, \quad \rho_{DM,EM} \in \{0.70, 0.80, 0.90\}.$$

These ranges reflect the empirical observation that global equity markets tend to be highly correlated, with correlations typically above 0.7 and often approaching 0.9 during periods of market

stress. For equity–bond relationships, I assume a common correlation parameter

$$\rho_{eq,bond} \in \{-0.10,\ 0.00,\ 0.10,\ 0.20\}.$$

Again, this range spans the historically observed variation in equity–bond correlation, which can shift from mildly negative during risk-off regimes to positive during inflation-driven drawdowns. Combining these grids produces $3 \times 3 \times 3 \times 4 = 108$ candidate correlation matrices. For each correlation configuration, I construct the 4×4 correlation matrix[18].

$$\begin{pmatrix} 1 & \rho_{US,DM} & \rho_{US,EM} & \rho_{eq,bond} \\ \rho_{US,DM} & 1 & \rho_{DM,EM} & \rho_{eq,bond} \\ \rho_{US,EM} & \rho_{DM,EM} & 1 & \rho_{eq,bond} \\ \rho_{eq,bond} & \rho_{eq,bond} & \rho_{eq,bond} & 1 \end{pmatrix}.$$

For each valid correlation matrix ρ, I construct the covariance matrix using $\Sigma = D\rho D$, where $D = \mathrm{diag}(\sigma_1, \ldots, \sigma_N)$ is the diagonal matrix of VCMM median 10-year volatilities. The grid-approach stress-tests correlation assumptions and ensures results are robust to uncertainty in cross-asset correlation.

4.3 Portfolio Analytics

Appendix A contains the 10-year median expected returns and median volatilities that I use as optimization inputs. I proxy the risk-free rate using the VCMM 10-year median U.S. cash return, $r_f = 3.5\%$ and use the 10-year inflation forecast of 2.0%. To compare portfolio performance, I use a number of analytics. For any portfolio with weights w, expected return is $\mu_p = w^\top \mu$ and given a covariance matrix Σ, portfolio volatility is $\sigma_p = \sqrt{w^\top \Sigma w}$. The Sharpe Ratio is therefore

$$SR_p = \frac{\mu_p - r_f}{\sigma_p}.$$

Section 3 detailed proxies for individual asset downside risk, upside, and skewness using VCMM quantiles. At the portfolio level, I use weighted quantiles

$$\text{Downside}_p = \sum_i w_i(P50_i - P5_i), \quad \text{Upside}_p = \sum_i w_i(P95_i - P50_i), \quad \text{Skew}_p = \text{Upside}_p - \text{Downside}_p$$

to approximate portfolio downside risk and distribution asymmetry[19]. I also repeat the risk-aversion calibration exercise detailed in Section 3 across the correlation grid.

[18] Note, not all such matrices are necessarily valid. A valid correlation matrix must be positive semidefinite (PSD), which I assess by computing the eigenvalues of each candidate matrix and retain only those for which $\min(\lambda_i) \geq 0$ (in practice, I allow a small numerical tolerance). Out of the 108 candidate matrices, 104 satisfy this condition.

[19] Simulation-based portfolio distributions or variance-based risk metrics, like conditional VaR, are ideal measures of tail risk, but I don't have access to the required data. The proxies I construct are not quantiles mathematically, as generally $Q_p(\sum_{i=1}^n w_i X_i) \neq \sum_{i=1}^n w_i Q_p(X_i)$, but they do reflect portfolio-level tail risk.

4.4 Baseline Portfolio Selection

For each valid covariance matrix in the correlation grid, I compute the performance statistics above for each of the three baseline portfolios. Table 18 contains the resulting distributions of expected return, volatility, Sharpe ratio, real return, tail proxies, and implied risk aversion. The results show the following. First, the market-composition portfolio dominates the equal-weight and risk-parity variants across most performance measures, including expected return, Sharpe ratio, real return, coefficient of variation, and downside risk proxy. Second, the minimal dispersion and apparent insensitivity to uncertainty in cross-asset correlations is partially attributable to the modest non-U.S. allocation of 12%. Consequently, I adopt the market-composition portfolio as the reference baseline for subsequent portfolio optimization.

To obtain a representative covariance structure for subsequent analysis, I select the correlation matrix closest to the median case across the grid. Specifically, I choose the configuration minimizing the standardized distance to the median values of portfolio volatility, Sharpe ratio, and implied risk aversion. The resulting base-case correlation matrix is

$$\rho = \begin{pmatrix} 1.00 & 0.75 & 0.65 & 0.10 \\ 0.75 & 1.00 & 0.90 & 0.10 \\ 0.65 & 0.90 & 1.00 & 0.10 \\ 0.10 & 0.10 & 0.10 & 1.00 \end{pmatrix}.$$

Using this covariance matrix, the baseline market-composition portfolio produces

$$\mu_p = 4.85\%, \qquad \sigma_p = 9.77\%, \qquad SR_p = 0.138, \quad \mu_p^{\text{real}} = 2.85\%,$$

with an implied risk-aversion coefficient of $\gamma \approx 0.068$[20].

4.5 New Portfolio Construction

I now expand the asset universe to include global ex-U.S. bonds and construct a new portfolio. I use the VCMM 10-year median expected returns and volatilities, together with the assumed correlation structure above. Again, consistent with my analysis for section 2 (preserving the fixed-income role of the bond sleeve by minimizing FX-driven volatility) and the VCMM series *Global ex-U.S. aggregate bonds (hedged)*, I assume a 100% hedge ratio.

Fundamentally, there are two types of portfolios. The first (60/40) retains the top-level 60/40 equity/bond structure and varies allocations *within* the equity and bond sleeves. These portfolios are akin to a TVAA imposed on the investor's current allocation. The second (Global) permits full global reoptimization across all five asset classes, and are akin to a new strategic asset allocation (SAA). The 60/40 portfolios preserve the investor's baseline risk profile and are more plausibly

[20] Note, this estimate differs markedly from the 0.19 obtained earlier. This reflects a different asset universe, and thus different expected returns and covariances. Moreover, the first estimate stems from applying least squares to 3 assets, and is necessarily noisy.

implemented. Global portfolios can be interpreted as benchmarks for how far unconstrained optimization would push the allocation. I include them for comparison.

For specific portfolio construction frameworks, I use conventional approaches, including equal weight, risk parity, maximize expected return, global minimum variance, and mean-variance optimal (See Appendix F for mathematical details). Table 10 summarizes the approaches.

Table 10: Portfolio Construction Problems and Closed-Form solutions

Portfolio	Optimization Problem	Closed-Form solution
Equal Weight	Impose Equal Weights	$w^{EW} = \frac{1}{N}\mathbf{1}$
Max. Exp. Ret.	$\max_w w'\mu$ subject to $\mathbf{1}'w = 1$, $w_i \geq 0$	All weight on asset with highest μ_i (corner solution)
GMV	$\min_w w'\Sigma w$ s.t. $\mathbf{1}'w = 1$	$w^{GMV} = \dfrac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}$
MVO	$\max_w w'\mu - \frac{\gamma}{2}w'\Sigma w$ s.t. $\mathbf{1}'w = 1$	See (5)
Tangency portfolio	$\max_w \frac{w'(\mu - r_f\mathbf{1})}{\sqrt{w'\Sigma w}}$	$w^T \propto \Sigma^{-1}(\mu - r_f\mathbf{1})$
Risk parity	Equalize $RC_i = \frac{w_i(\Sigma w)_i}{\sigma_p}$ across assets	No general closed form, if Σ diagonal, $w_i \propto \sigma_i^{-1}$

With an additional asset, global ex-U.S. bonds, comes a requirement for another set of correlations. I retain the original correlations among U.S. equities, developed ex-U.S. equities, emerging markets equities, and U.S. aggregate bonds from the base-case matrix, and then vary two additional correlation parameters. I assume a common correlation between global ex-U.S. bonds and all equity sleeves,

$$\rho_{\text{GXUS bond, equity}} = \{-0.10,\ 0.00,\ 0.10,\ 0.20\}.$$

and a common correlation between U.S. aggregate bonds and global ex-U.S. bonds,

$$\rho_{\text{bond, bond}} = \{0.50,\ 0.60,\ 0.70,\ 0.80\}.$$

The resulting five-asset correlation matrix preserves the original equity-equity and U.S. bond-equity structure and permits uncertainty around the new foreign bond sleeve.

4.6 Results

Table 19 shows results for the new portfolio construction exercise, using the full suite of portfolio construction methods and the 16 covariance matrices. The baseline portfolio, with an expected return of 4.846%, volatility of 9.77%, and Sharpe ratio of 0.138, remains the reference point. Table 11 shows allocations. I observe the following.

First, results are driven primarily by VCMM inputs. Within the 10-year VCMM forecasts, the highest expected return in the five-asset universe is developed ex-U.S. equities, at 5.9%, while the lowest volatility belongs to global ex-U.S. bonds, at 5.0%. It is thus entirely unsurprising that the optimization algorithms, especially MVO and GMV, heavily favor these two assets.

Second, consider portfolios that retain the top-level 60/40 equity/bond split. The 60/40 equal-weight portfolio allocates 20% to each of the five assets. It delivers an expected return of 4.92%, volatility of 11.29%, and a Sharpe ratio of 0.126. Volatility rises because equal weighting overweights the higher-volatility equity assets, especially EM and developed ex-U.S. equities.

The 60/40 risk-parity portfolio allocates approximately 24.7% to U.S. equities, 20.6% to developed ex-U.S. equities, 14.7% to EM equities, 17.7% to U.S. bonds, and 22.3% to global bonds. It generates an expected return of 4.97%, volatility of 10.73%, and a Sharpe ratio of 0.137. Risk parity places more weight on bonds and low-volatility assets, which improves the balance of risk contributions. As a result, its Sharpe ratio is close (marginally lower) to the baseline's.

The 60/40 GMV portfolio puts 50.4% in U.S. equities, 9.6% in developed ex-U.S. equities, 0% in EM equities, 7.4% in U.S. bonds, and 32.6% in global bonds. It achieves an expected return of 5.01%, volatility of 9.39%, and a Sharpe ratio of 0.161. This is an improvement over the baseline on all three dimensions. The allocation eliminates EM, increases global bonds substantially, and concentrates the equity sleeve more heavily in the U.S.

The 60/40 MVO portfolio places 0% in U.S. equities, 60% in developed ex-U.S. equities, 0% in EM equities, 0% in U.S. bonds, and 40% in global bonds. It delivers an expected return of 5.54%, volatility of 11.20%, and a Sharpe ratio of 0.182. This is again higher than the baseline, but it is a classic corner solution. The optimizer simply chooses the highest-return asset in each sleeve, namely developed ex-U.S. equities for the equity sleeve, and global ex-U.S. bonds for the bond sleeve.

The 60/40 Max Expected Return portfolio produces the same allocation and same performance statistics as the 60/40 MVO portfolio. This occurs because developed ex-U.S. equities have the highest expected return among equities and global ex-U.S. bonds have the highest expected return among bonds. The portfolio therefore collapses to 60% developed ex-U.S. equities and 40% global bonds.

Third, consider the global portfolios. Global equal weight is the same as 60/40 equal weight in this setup, since all five assets again receive 20%. Global risk parity allocates approximately 12.7% to U.S. equities, 10.6% to developed ex-U.S. equities, 7.5% to EM equities, 30.6% to U.S. bonds, and 38.6% to global bonds. It generates an expected return of 4.90%, volatility of 6.63%, and a Sharpe ratio of 0.211. This is a meaningful improvement in risk-adjusted performance relative to the baseline. The reason is simple: global risk parity loads much more heavily into low-volatility bond assets, so portfolio volatility declines substantially.

Global GMV allocates 7.1% to U.S. equities, 0.6% to developed ex-U.S. equities, 0% to EM equities, 14.7% to U.S. bonds, and 75.3% to global bonds. It produces an expected return of 4.93%, volatility of 4.74%, and a Sharpe ratio of 0.307, the highest Sharpe ratio in the set. As

expected, it loads heavily on the lowest volatility asset, which in this universe is global ex-U.S. bonds. However, this is not economically realistic. The portfolio is far too bond-heavy, and thus too exposed to interest rate risk, credit risk, and inflation rate cycles.

Global MVO allocates 100% to developed ex-U.S. equities. It generates an expected return of 5.9%, volatility of 18.2%, and a Sharpe ratio of 0.132. This is another classic corner solution, and is a product of high dispersion in expected returns with no structural constraints.

Table 11: Median Portfolio Weights Across Correlation Grid

Portfolio	US Eq.	DM ex-US Eq.	EM Eq.	US Agg. Bond	Global ex-US Bond
60/40 Portfolios					
Equal Weight	20.0%	20.0%	20.0%	20.0%	20.0%
GMV	50.4%	9.6%	0.0%	7.4%	32.6%
MVO	0.0%	60.0%	0.0%	0.0%	40.0%
Max ER	0.0%	60.0%	0.0%	0.0%	40.0%
Risk Parity	24.7%	20.6%	14.7%	17.7%	22.3%
Global Portfolios					
Baseline	48.0%	8.8%	3.2%	40.0%	0.0%
Equal Weight	20.0%	20.0%	20.0%	20.0%	20.0%
GMV	7.1%	0.6%	0.0%	14.7%	75.3%
MVO	0.0%	100.0%	0.0%	0.0%	0.0%
Risk Parity	12.7%	10.6%	7.5%	30.6%	38.6%

4.7 Recommended Portfolio

I identify two serious candidates for a new portfolio, 60/40 GMV and 60/40 risk parity. Both preserve the policy structure and thus are closer to a TVAA than to a new SAA. Both diversify internationally, and both have Sharpe ratios close (0.161 and 0.137, respectively) to the baseline. I favor the 60/40 GMV portfolio, as it delivers nearly uniformly better performance than the baseline.

Table 12: Baseline vs. 60/40 GMV Portfolio

Metric	Baseline	60/40 GMV
Expected Return	4.846%	**5.010%**
Volatility	9.766%	**9.385%**
Sharpe Ratio	0.138	**0.161**
Real Return	2.846%	**3.010%**
Coeff. Var.	2.015	**1.873**
Left Tail Proxy	5.593%	**5.712%**
Upside Proxy	6.013%	**6.078%**
Skew Proxy	**0.419%**	0.366%

The Sharpe ratio rises by roughly 17%, while volatility declines and expected return increases, reflecting the benefits of diversification, specifically higher expected returns on foreign equities and a global bond sleeve that offers lower volatility and marginally higher expected return than U.S. aggregate bonds.

I outline several other reasons to favor the 60/40 GMV portfolio. First, the volatility and Sharpe ratio of 60/40 GMV barely changes across the correlation grid. The minimum Sharpe ratio is 0.155, the median is 0.161, and the maximum is 0.170. This indicates that the GMV allocation is robust to the correlation assumptions[21].

Second, consider left-tail risk. Global GMV is lowest at 2.4, followed by global risk parity at 3.8, 60/40 GMV at 5.7, and the baseline at 5.6. The very low-volatility portfolios do dramatically reduce downside risk, but they do so only by becoming too bond-heavy. By contrast, 60/40 GMV keeps left-tail risk approximately unchanged relative to the baseline while still increasing expected return.

Third, consider the required reallocations relative to the baseline. Table 13 compares the portfolio weights and shows required reallocations to get from the baseline to 60/40 GMV.

Table 13: Baseline vs. Recommended 60/40 GMV Portfolio

Asset	Baseline	60/40 GMV	Change
U.S. equities	48.0%	50.4%	+2.4%
DM ex-U.S. equities	8.8%	9.6%	+0.8%
EM equities	3.2%	0.0%	-3.2%
U.S. bonds	40.0%	7.4%	**-32.6%**
Global ex-U.S. bonds	0.0%	32.6%	**+32.6%**

[21] The median weights also imply a base-case correlation matrix that is broadly consistent with historical intuition: equities are highly correlated globally, bonds are weakly correlated with equities, and global bonds are moderately correlated with U.S. bonds. The recommendation doesn't embed large negative-correlation assumptions.

Reallocating requires selling a large portion of the U.S. bond sleeve and replace it with hedged global bonds. The equity changes are comparatively modest. The investor has to sell the small EM allocation and reallocate that capital mostly to U.S. equities, with a smaller increase in developed ex-U.S. equities. Transaction costs naturally depend on the implementation vehicle, but in practice the largest adjustment is within fairly liquid bond markets. Bid-ask spreads, small commissions, and market impact should therefore be manageable, particularly if the reallocation is implemented using broad funds or ETFs.

A final consideration for the tax aware investor is tax liabilities. Selling a large U.S. bond position may trigger capital gains taxes if the portfolio has appreciated, thus the shift can be costly in a taxable account. However, I argue this isn't a unique disadvantage. Any portfolio with a higher Sharpe ratio than the baseline will have some reallocation cost, which in practice can be mitigated by spreading the transition over time, directing new contributions into global bonds, and taking advantage of tax-loss harvesting where available. A tax-aware portfolio implementation would phase in the changes over time, prioritize new contributions, and avoid realizing capital gains unnecessarily.

A VCMM Forecasts

Table 14: VCMM Distribution Summary: 10-year and 30-year Horizons

Horizon	Statistic	U.S. Eq.	DM ex-U.S. Eq.	EM Eq.	U.S. Agg. Bonds	U.S. Cash	U.S. Inflation
10Y	5th Percentile	-3.5	-1.5	-5.5	3.1	1.7	0.6
10Y	25th Percentile	1.5	2.9	0.1	4.0	2.8	1.5
10Y	50th Percentile	4.9	5.9	4.2	4.6	3.5	2.0
10Y	75th Percentile	8.4	9.1	8.4	5.2	4.3	2.6
10Y	95th Percentile	13.9	13.9	15.1	6.2	5.4	3.5
10Y	Median Volatility	15.2	18.2	25.6	6.3	1.1	1.7
30Y	5th Percentile	-0.2	3.0	-0.1	3.5	1.8	0.9
30Y	25th Percentile	3.4	5.4	3.7	4.2	2.7	1.6
30Y	50th Percentile	5.6	7.0	6.1	4.6	3.3	2.0
30Y	75th Percentile	7.9	8.6	8.6	5.1	3.8	2.4
30Y	95th Percentile	11.6	11.1	12.8	5.8	4.9	3.1
30Y	Median Volatility	17.0	20.1	27.9	6.5	1.5	1.9

Table 15: VCMM Derived Metrics: 10-year and 30-year Horizons

Horizon	Statistic	U.S. Eq.	DM ex-U.S. Eq.	EM Eq.	U.S. Agg. Bonds	U.S. Cash	U.S. Inflation
10Y	Excess Return	1.4	2.4	0.7	1.1	0.0	-1.5
10Y	Sharpe Ratio	0.092	0.132	0.027	0.175	0.000	-0.882
10Y	Downside Risk	8.4	7.4	9.7	1.5	1.8	1.4
10Y	Upside	9.0	8.0	10.9	1.6	1.9	1.5
10Y	Skew Proxy	0.6	0.6	1.2	0.1	0.1	0.1
30Y	Excess Return	2.3	3.7	2.8	1.3	0.0	-1.3
30Y	Sharpe Ratio	0.135	0.184	0.100	0.200	0.000	-0.684
30Y	Downside risk	5.8	4.0	6.2	1.1	1.5	1.1
30Y	Upside	6.0	4.1	6.7	1.2	1.6	1.1
30Y	Skew Proxy	0.2	0.1	0.5	0.1	0.1	0.0

B Mathematical Framework for Currency Hedging

Let S_t represent the exchange rate for home currency (USD) to foreign currency, denoted F. Thus S_t represents the USD price of one unit of foreign currency:

$$S_t = \frac{\text{USD}}{\text{1 unit of foreign currency}}.$$

Let P_t^F be the price of the foreign bond at time t, measured in foreign currency. Let $C_{t,t+1}^F$ be the bond's cash flows received between times t and $t+1$, again measured in foreign currency.

The gross return in foreign currency is

$$1 + R_{t,t+1}^F = \frac{P_{t+1}^F + C_{t,t+1}^F}{P_t^F}.$$

A U.S.-based investor ultimately converts portfolio wealth to USD. The initial USD value of the position at time t is

$$P_t^{\text{USD}} = S_t P_t^F,$$

and the final USD value at $t+1$ is

$$P_{t+1}^{\text{USD}} + C_{t,t+1}^{\text{USD}} = S_{t+1}(P_{t+1}^F + C_{t,t+1}^F).$$

The gross USD return is therefore

$$1 + R_{t,t+1}^{\text{USD}} = \frac{S_{t+1}(P_{t+1}^F + C_{t,t+1}^F)}{S_t P_t^F}.$$

Rearranging gives

$$\begin{aligned} 1 + R_{t,t+1}^{\text{USD}} &= \frac{P_{t+1}^F + C_{t,t+1}^F}{P_t^F} \cdot \frac{S_{t+1}}{S_t} \\ &= (1 + R_{t,t+1}^F)(1 + R_{t,t+1}^{FX}), \end{aligned}$$

where the gross FX return is

$$1 + R_{t,t+1}^{\text{FX}} = \frac{S_{t+1}}{S_t}.$$

Note the following. First, the equation is multiplicative in the foreign bond return and the FX return. If the bond appreciates in foreign-currency terms and the foreign currency appreciates against the USD, then the FX return amplifies the USD bond return. If the foreign currency depreciates, the FX return dampens the USD return. Second, the foreign bond portfolio has exposure to two sources of risk: one from the bond position itself (interest-rate risk and credit risk), and one from FX risk. The sources of uncertainty are the future spot rate S_{t+1} and the future foreign bond price P_{t+1}^F. All other terms are either known at time t or treated as deterministic (e.g.

intervening bond cash flows).

Hedging in this context refers to removing uncertainty arising from the future spot rate. Currency hedging keeps the foreign bond exposure but mitigates risk from future currency conversion. The principal hedging instrument is an FX forward, in which the USD investor agrees at time t to sell foreign currency at $t+1$ at a predetermined forward exchange rate $F_{t,t+1}$.

The future USD value of the bond position's proceeds is thus

$$F_{t,t+1}(P^F_{t+1} + C^F_{t,t+1}),$$

and the USD return with full hedging is

$$1 + R^{\text{USD}}_{t,t+1} = \frac{P^F_{t+1} + C^F_{t,t+1}}{P^F_t} \cdot \frac{F_{t,t+1}}{S_t}.$$

Now the only random term, and thus the only source of risk, is the time $t+1$ foreign bond price.

Currency risk is eliminated. The question is now how much foreign currency exposure the USD investor should hedge. It is not obvious that eliminating all FX risk is always desirable. Excess returns are compensation for bearing systematic risk, and as stated earlier, a positive FX return can increase the USD return on the position.

A hedge ratio measures how much of that FX exposure is hedged, in this case offset using forward contracts. Let H^F_t be the amount of foreign currency that the U.S. investor sells forward. Then the hedge ratio h_t is

$$h_t = \frac{H^F_t}{P^F_t}.$$

As prices are non-negative, h_t is naturally bounded between 0 and 1. When $h_t = 0$, FX risk is not hedged at all, and when $h_t = 1$, it is fully hedged. For now, leave h_t as an unspecified fraction such that at time $t+1$, $h_t(P^F_{t+1} + C^F_{t,t+1})$ is converted to USD at the forward rate $F_{t,t+1}$ and $(1-h_t)(P^F_{t+1} + C^F_{t,t+1})$ is converted to USD at the spot rate S_{t+1}.

Thus the USD portfolio value at $t+1$ is

$$F_{t,t+1}h_t(P^F_{t+1} + C^F_{t,t+1}) + S_{t+1}(1-h_t)(P^F_{t+1} + C^F_{t,t+1}),$$

with the same initial investment in USD of $S_t P^F_t$. The gross USD return is thus

$$\begin{aligned}
1 + R^{\text{USD}}_{t,t+1} &= \frac{(P^F_{t+1} + C^F_{t,t+1})[F_{t,t+1}h_t + S_{t+1}(1-h_t)]}{S_t P^F_t} \\
&= \frac{P^F_{t+1} + C^F_{t,t+1}}{P^F_t} \frac{[F_{t,t+1}h_t + S_{t+1}(1-h_t)]}{S_t} \\
&= (1 + R^F_{t,t+1}) \frac{[F_{t,t+1}h_t + S_{t+1}(1-h_t)]}{S_t} \\
&= (1 + R^F_{t,t+1}) \left[h_t \frac{F_{t,t+1}}{S_t} + (1-h_t)(1 + R^{\text{FX}}_{t,t+1}) \right].
\end{aligned}$$

The forward rate $F_{t,t+1}$ reflects the interest-rate differential between the two countries, a result known as covered interest parity. Let i_t^{USD} be the USD risk-free rate from t to $t+1$, and let i_t^F be the foreign-currency risk-free rate over the same period. By no arbitrage,

$$\frac{F_{t,t+1}}{S_t} = \frac{1 + i_t^{\text{USD}}}{1 + i_t^F}.$$

Intuitively, the forward rate must adjust so that it is not possible to borrow in the currency of the country with the lower rate, convert at the spot rate, lend in the higher-rate currency, and lock in a riskless profit by using a forward contract to convert the proceeds back into the original currency.

This interest-rate differential is the source of hedging carry. If $i_t^{\text{USD}} > i_t^F$, then $\frac{F_{t,t+1}}{S_t} > 1$, and hedging amplifies the USD return relative to the foreign-currency return. Conversely if $i_t^{\text{USD}} < i_t^F$, then hedging reduces it. This is why a low-yield foreign bond can still generate attractive USD returns when the foreign bond position is fully hedged.

CIP generally holds, but deviations are possible. Basis risk arises when the forward rate deviates from the CIP benchmark. Let the cross-currency basis be b_t. Then

$$\frac{F_{t,t+1}}{S_t} = \frac{1 + i_t^{\text{USD}}}{1 + i_t^F}(1 + b_t).$$

When $b_t \neq 0$, basis risk filters into portfolio moments through the ratio of forward to spot rates. If the basis changes, future hedge rollovers occur at different forward rates, which affects realized portfolio returns.

Finally, I turn to portfolio volatility. Recall that the random quantities are P_{t+1}^F and S_{t+1}, and the deterministic quantities are P_t^F, S_t, $F_{t,t+1}$, and h_t. Thus the foreign bond return is random through the next-period bond price, and the FX return is random through the future spot rate.

First, consider expected returns:

$$\begin{aligned}
E_t(1 + R_{t,t+1}^{\text{USD}}) &= E_t\left[(1 + R_{t,t+1}^F)\left(h_t \frac{F_{t,t+1}}{S_t} + (1 - h_t)(1 + R_{t,t+1}^{\text{FX}})\right)\right] \\
&= h_t \frac{F_{t,t+1}}{S_t} E_t(1 + R_{t,t+1}^F) + (1 - h_t) E_t\left[(1 + R_{t,t+1}^F)(1 + R_{t,t+1}^{\text{FX}})\right] \\
&= h_t \frac{1 + i_t^{\text{USD}}}{1 + i_t^F} E_t(1 + R_{t,t+1}^F) \\
&+ (1 - h_t)\left(\text{Cov}_t(1 + R_{t,t+1}^F,\, 1 + R_{t,t+1}^{\text{FX}}) + E_t(1 + R_{t,t+1}^F) E_t(1 + R_{t,t+1}^{\text{FX}})\right).
\end{aligned}$$

The results discussed above now follow cleanly from this expression. When $h_t = 1$ and the position is fully hedged, expected return is purely a function of expected return on the bond position and the hedging carry. When $h_t = 0$ and the position is fully unhedged, the expected USD gross return depends on both expected FX returns and the covariance between the foreign bond return and the FX return. If that covariance term is positive (negative), expected USD gross

return is higher (lower).

Second, compute conditional variance:

$$R^{\text{USD}}_{t,t+1} = (1 + R^F_{t,t+1}) \left[h_t \frac{F_{t,t+1}}{S_t} + (1 - h_t)(1 + R^{\text{FX}}_{t,t+1}) \right] - 1.$$

Thus

$$\begin{aligned}
\text{Var}_t(R^{\text{USD}}_{t,t+1}) = &\left(h_t \frac{F_{t,t+1}}{S_t} \right)^2 \text{Var}_t(R^F_{t,t+1}) \\
&+ (1 - h_t)^2 \, \text{Var}_t \left[(1 + R^F_{t,t+1})(1 + R^{\text{FX}}_{t,t+1}) \right] \\
&+ 2(1 - h_t) h_t \frac{F_{t,t+1}}{S_t} \text{Cov}_t \left[1 + R^F_{t,t+1},\, (1 + R^F_{t,t+1})(1 + R^{\text{FX}}_{t,t+1}) \right].
\end{aligned}$$

Expanding the product terms gives

$$\begin{aligned}
\text{Var}_t(R^{\text{USD}}_{t,t+1}) = &\left(h_t \frac{F_{t,t+1}}{S_t} \right)^2 \text{Var}_t(R^F_{t,t+1}) \\
&+ (1 - h_t)^2 \, \text{Var}_t \left[R^F_{t,t+1} + R^{\text{FX}}_{t,t+1} + R^F_{t,t+1} R^{\text{FX}}_{t,t+1} \right] \\
&+ 2(1 - h_t) h_t \frac{F_{t,t+1}}{S_t} \text{Cov}_t \left[R^F_{t,t+1},\, R^F_{t,t+1} + R^{\text{FX}}_{t,t+1} + R^F_{t,t+1} R^{\text{FX}}_{t,t+1} \right].
\end{aligned}$$

Expanding variance and covariance terms further yields

$$\begin{aligned}
\text{Var}_t(R^{\text{USD}}_{t,t+1}) = &\left(h_t \frac{F_{t,t+1}}{S_t} \right)^2 \text{Var}_t(R^F_{t,t+1}) \\
&+ (1 - h_t)^2 \Bigg(\text{Var}_t(R^F_{t,t+1}) + \text{Var}_t(R^{\text{FX}}_{t,t+1}) + \text{Var}_t(R^F_{t,t+1} R^{\text{FX}}_{t,t+1}) \\
&\qquad + 2\text{Cov}_t(R^F_{t,t+1}, R^{\text{FX}}_{t,t+1}) + 2\text{Cov}_t(R^F_{t,t+1}, R^F_{t,t+1} R^{\text{FX}}_{t,t+1}) + 2\text{Cov}_t(R^{\text{FX}}_{t,t+1}, R^F_{t,t+1} R^{\text{FX}}_{t,t+1}) \Bigg) \\
&+ 2(1 - h_t) h_t \frac{F_{t,t+1}}{S_t} \Bigg(\text{Var}_t(R^F_{t,t+1}) + \text{Cov}_t(R^F_{t,t+1}, R^{\text{FX}}_{t,t+1}) + \text{Cov}_t(R^F_{t,t+1}, R^F_{t,t+1} R^{\text{FX}}_{t,t+1}) \Bigg).
\end{aligned}$$

Collecting terms gives

$$\begin{aligned}
\text{Var}_t(R^{\text{USD}}_{t,t+1}) = &\left[\left(h_t \frac{F_{t,t+1}}{S_t} \right)^2 + 2(1 - h_t) h_t \frac{F_{t,t+1}}{S_t} + (1 - h_t)^2 \right] \text{Var}_t(R^F_{t,t+1}) \\
&+ (1 - h_t)^2 \text{Var}_t(R^{\text{FX}}_{t,t+1}) + (1 - h_t)^2 \text{Var}_t(R^F_{t,t+1} R^{\text{FX}}_{t,t+1}) \\
&+ 2(1 - h_t) \left[h_t \frac{F_{t,t+1}}{S_t} + (1 - h_t) \right] \text{Cov}_t(R^F_{t,t+1}, R^{\text{FX}}_{t,t+1}) \\
&+ 2(1 - h_t) \left[h_t \frac{F_{t,t+1}}{S_t} + (1 - h_t) \right] \text{Cov}_t(R^F_{t,t+1}, R^F_{t,t+1} R^{\text{FX}}_{t,t+1}) \\
&+ 2(1 - h_t)^2 \text{Cov}_t(R^{\text{FX}}_{t,t+1}, R^F_{t,t+1} R^{\text{FX}}_{t,t+1}).
\end{aligned}$$

This final expression is somewhat unwieldy, so I use a first-order linear approximation to drop the product terms $R^F_{t,t+1}R^{\text{FX}}_{t,t+1}$ and obtain

$$\begin{aligned}\text{Var}_t(R^{\text{USD}}_{t,t+1}) \approx &\left[\left(h_t\frac{F_{t,t+1}}{S_t}\right)^2 + 2(1-h_t)h_t\frac{F_{t,t+1}}{S_t} + (1-h_t)^2\right]\text{Var}_t(R^F_{t,t+1}) \\ &+ (1-h_t)^2\text{Var}_t(R^{\text{FX}}_{t,t+1}) \\ &+ 2(1-h_t)\left[h_t\frac{F_{t,t+1}}{S_t} + (1-h_t)\right]\text{Cov}_t(R^F_{t,t+1}, R^{\text{FX}}_{t,t+1}),\end{aligned}$$

which neatly decomposes USD return variance into foreign bond variance, FX return variance, and the covariance between the two.

C VCMM Return–Volatility Plots and Sharpe Ratios

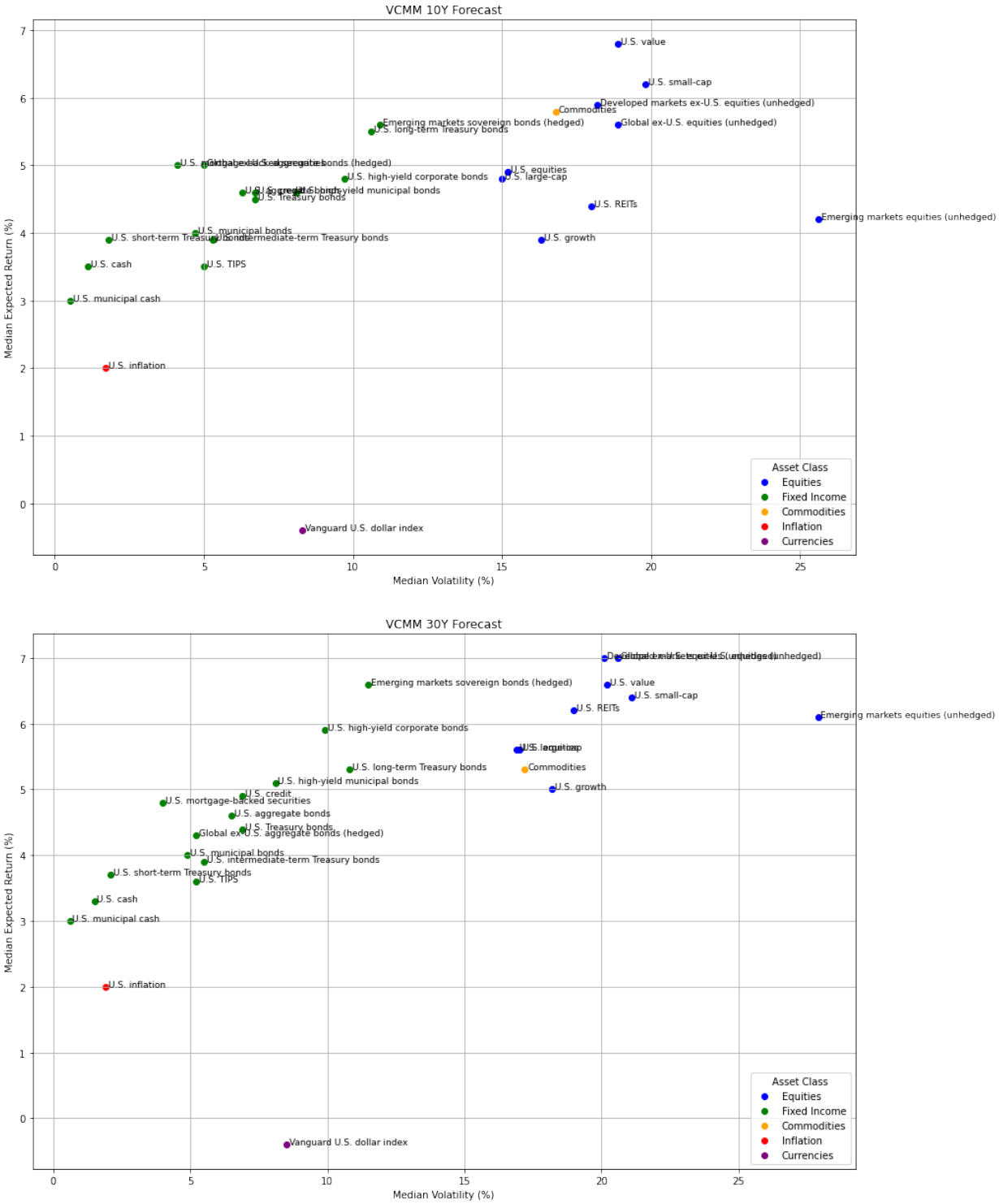

Figure 1: VCMM Forecasts, 10-year (top) and 30-year (bottom).

Table 16: Sharpe Ratios by Asset, 10-year vs. 30-year

Asset	Class	SR 10Y	SR 30Y	SR Diff.
U.S. municipal cash	Fixed Income	-1.000000	-0.500000	0.500000
U.S. inflation	Inflation	-0.882353	-0.684211	0.198142
U.S. high-yield corporate bonds	Fixed Income	0.134021	0.262626	0.128606
U.S. REITs	Equities	0.050000	0.152632	0.102632
Emerging markets sovereign bonds (hedged)	Fixed Income	0.192661	0.286957	0.094296
U.S. high-yield municipal bonds	Fixed Income	0.135802	0.222222	0.086420
Emerging markets equities (unhedged)	Equities	0.027344	0.100358	0.073015
U.S. growth	Equities	0.024540	0.093407	0.068867
Global ex-U.S. equities (unhedged)	Equities	0.111111	0.179612	0.068501
U.S. credit	Fixed Income	0.164179	0.231884	0.067705
U.S. TIPS	Fixed Income	0.000000	0.057692	0.057692
Developed markets ex-U.S. equities (unhedged)	Equities	0.131868	0.184080	0.052211
U.S. large-cap	Equities	0.086667	0.136095	0.049428
U.S. equities	Equities	0.092105	0.135294	0.043189
U.S. municipal bonds	Fixed Income	0.106383	0.142857	0.036474
Vanguard U.S. dollar index	Currencies	-0.469880	-0.435294	0.034585
U.S. intermediate-term Treasury bonds	Fixed Income	0.075472	0.109091	0.033619
U.S. aggregate bonds	Fixed Income	0.174603	0.200000	0.025397
U.S. small-cap	Equities	0.136364	0.146919	0.010556
U.S. Treasury bonds	Fixed Income	0.149254	0.159420	0.010167
U.S. mortgage-backed securities	Fixed Income	0.365854	0.375000	0.009146
U.S. cash	Fixed Income	0.000000	0.000000	0.000000
U.S. long-term Treasury bonds	Fixed Income	0.188679	0.185185	-0.003494
U.S. value	Equities	0.174603	0.163366	-0.011237
Commodities	Commodities	0.136905	0.116279	-0.020626
U.S. short-term Treasury bonds	Fixed Income	0.222222	0.190476	-0.031746
Global ex-U.S. aggregate bonds (hedged)	Fixed Income	0.300000	0.192308	-0.107692

Table 17: Average Change in Sharpe Ratios by Asset Class

Class	Average SR Diff.
Commodities	-0.020626
Currencies	0.034585
Equities	0.050796
Fixed Income	0.060439
Inflation	0.198142

D Calibration of Investor Risk Aversion from Policy Portfolio

This appendix gives the explicit mathematical and econometric setup used to calibrate the investor's risk-aversion parameter γ from the policy portfolio. Consider the standard mean-variance optimization problem

$$\max_{w} \ \mu' w - \frac{\gamma}{2} w' \Sigma w$$

subject to

$$\mathbf{1}' w = 1,$$

where:

- w is the vector of portfolio weights,
- μ is the vector of expected returns,
- Σ is the covariance matrix of returns,
- $\gamma > 0$ is the coefficient of risk aversion.

The Lagrangian is

$$\mathcal{L}(w, \lambda) = \mu' w - \frac{\gamma}{2} w' \Sigma w + \lambda(1 - \mathbf{1}' w).$$

The first-order condition with respect to w is

$$\mu - \gamma \Sigma w - \lambda \mathbf{1} = 0,$$

or equivalently,

$$\mu = \gamma \Sigma w + \lambda \mathbf{1}. \tag{1}$$

To calibrate γ, I treat the policy portfolio w^P as the observed optimal allocation under the 30-year, or steady-state, return forecasts. Substituting w^P into (1) gives

$$\mu = \gamma \Sigma w^P + \lambda \mathbf{1}. \tag{2}$$

This is the key reverse-optimization relationship. Expected returns should be proportional to marginal risk contributions, up to an intercept term λ coming from the budget constraint. Define

$$s = \Sigma w^P.$$

The vector s collects the marginal contributions of each asset to portfolio variance, thus (2) can be written asset by asset as

$$\mu_i = \gamma s_i + \lambda, \qquad i = 1, 2, 3.$$

The calibration problem can be expressed as a cross-sectional linear regression of expected returns on marginal risk contributions. To estimate γ and λ, I use least squares. Stack the three

equations as

$$\mu = X\beta + \varepsilon,$$

where

$$\beta = \begin{pmatrix} \gamma \\ \lambda \end{pmatrix}, \qquad X = \begin{bmatrix} s_1 & 1 \\ s_2 & 1 \\ s_3 & 1 \end{bmatrix}, \qquad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix}.$$

The OLS estimator is

$$\begin{pmatrix} \hat{\gamma} \\ \hat{\lambda} \end{pmatrix} = (X'X)^{-1}X'\mu. \tag{3}$$

E VTWAX Allocation

Regions **Markets**

	Regions	VTWAX
●	Emerging Markets	10.30%
●	Europe	14.60%
●	Pacific	10.20%
●	Middle East	0.30%
●	North America	64.40%
●	Other	0.20%

Weighted equity exposures exclude any temporary cash investments and equity index futures. Some short-term fixed income securities are classified as cash and are excluded from the weighted bond exposures.

Figure 2: Regional composition of Vanguard Total World Stock Index Fund.

F Portfolio Optimization Problems and Closed Form Solutions

This appendix presents the portfolio construction problems used throughout the analysis and their closed form solutions where available.

F.1 Setup and Notation

Let there be N risky assets. Let

$$w = \Big(w_1 \dots w_N\Big)^\top$$

be the vector of portfolio weights,

$$\mu = \Big(\mu_1 \dots \mu_N\Big)^\top$$

the vector of expected returns, and

$$\Sigma \in \mathbb{R}^{N \times N}$$

the covariance matrix of returns. Let $\mathbf{1}$ denote the $N \times 1$ vector of ones. Portfolio expected return and variance are

$$\mu_p = w'\mu, \qquad \sigma_p^2 = w'\Sigma w.$$

Unless otherwise stated, the budget constraint is

$$\mathbf{1}'w = 1.$$

In some applications I also impose long-only constraints

$$w_i \geq 0 \qquad \text{for all } i = 1, \dots, N,$$

in which case closed-form solutions generally do not exist and the problem must be solved numerically.

F.2 Equal-Weight

The equal-weight portfolio assigns identical weight to each asset:

$$w_i^{EW} = \frac{1}{N}, \qquad i = 1, \dots, N.$$

Equivalently,

$$w^{EW} = \frac{1}{N}\mathbf{1}.$$

This portfolio does not depend on expected returns, volatilities, or correlations and is often used as a naive diversification benchmark.

F.3 Maximum Expected Return

The maximum expected return portfolio solves

$$\max_{w} \; w'\mu$$

subject to

$$\mathbf{1}'w = 1, \qquad w_i \geq 0.$$

Under long-only constraints, the solution is a corner portfolio: allocate all weight to the asset with the highest expected return. Let

$$j^* = \arg\max_{j} \mu_j.$$

Then

$$w_i^{MaxER} = \begin{cases} 1, & i = j^*, \\ 0, & i \neq j^*. \end{cases}$$

Absent long-only constraints, the problem is ill-posed because leverage can be scaled without bound.

F.4 Global Minimum Variance

The global minimum variance (GMV) portfolio solves

$$\min_{w} \; w'\Sigma w$$

subject to

$$\mathbf{1}'w = 1.$$

Without constraints, a closed form solution exists. The Lagrangian is

$$\mathcal{L}(w, \lambda) = w'\Sigma w + \lambda(1 - \mathbf{1}'w).$$

The first-order condition is

$$2\Sigma w - \lambda \mathbf{1} = 0,$$

which implies

$$w = \frac{\lambda}{2}\Sigma^{-1}\mathbf{1}.$$

Imposing the budget constraint $\mathbf{1}'w = 1$ yields

$$\frac{\lambda}{2} = \frac{1}{\mathbf{1}'\Sigma^{-1}\mathbf{1}},$$

so the GMV portfolio is

$$w^{GMV} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}. \tag{4}$$

If $w_i \geq 0$, the solution generally has no closed form and must be computed numerically.

F.5 Mean-Variance Optimal

The mean-variance optimal (MVO) portfolio solves

$$\max_w \; w'\mu - \frac{\gamma}{2} w'\Sigma w$$

subject to

$$\mathbf{1}'w = 1,$$

where $\gamma > 0$ is the investor's coefficient of risk aversion. Equivalently, the problem can be written as

$$\min_w \; \frac{\gamma}{2} w'\Sigma w - w'\mu.$$

Without constraints, there is a closed form solution. The Lagrangian is

$$\mathcal{L}(w, \lambda) = w'\mu - \frac{\gamma}{2} w'\Sigma w + \lambda(1 - \mathbf{1}'w).$$

The first-order condition is

$$\mu - \gamma\Sigma w - \lambda\mathbf{1} = 0,$$

or

$$\gamma\Sigma w = \mu - \lambda\mathbf{1}.$$

Thus

$$w = \frac{1}{\gamma}\Sigma^{-1}(\mu - \lambda\mathbf{1}).$$

Imposing the budget constraint gives

$$\mathbf{1}'w = \frac{1}{\gamma}\left(\mathbf{1}'\Sigma^{-1}\mu - \lambda\mathbf{1}'\Sigma^{-1}\mathbf{1}\right) = 1.$$

Hence

$$\lambda = \frac{\mathbf{1}'\Sigma^{-1}\mu - \gamma}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}.$$

Substituting back yields the closed-form MVO portfolio:

$$w^{MVO} = \frac{1}{\gamma}\Sigma^{-1}\left(\mu - \frac{\mathbf{1}'\Sigma^{-1}\mu - \gamma}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}\mathbf{1}\right). \tag{5}$$

If $w_i \geq 0$, the unconstrained solution in (5) may be infeasible, and the constrained problem must be solved numerically.

F.6 Mean-Variance with Risk-Free Asset

If a risk-free asset with return r_f is available, define excess returns

$$\mu^e = \mu - r_f \mathbf{1}.$$

Then the tangency portfolio solves

$$\max_w \frac{w'\mu^e}{\sqrt{w'\Sigma w}}.$$

Equivalently, under a normalization of portfolio scale, the tangency portfolio is proportional to

$$w^T \propto \Sigma^{-1}\mu^e. \tag{6}$$

If weights are normalized to sum to one, then

$$w^T = \frac{\Sigma^{-1}\mu^e}{\mathbf{1}'\Sigma^{-1}\mu^e}. \tag{7}$$

This is distinct from the fully invested risky-asset MVO problem in (5) because the latter embeds the budget constraint directly in the risky portfolio.

F.7 Risk Parity

Risk parity chooses weights so that each asset contributes equally to total portfolio risk.

Portfolio volatility is

$$\sigma_p = \sqrt{w'\Sigma w}.$$

The marginal contribution of asset i to volatility is

$$\frac{\partial \sigma_p}{\partial w_i} = \frac{(\Sigma w)_i}{\sigma_p}.$$

The total risk contribution of asset i is

$$RC_i = w_i \frac{\partial \sigma_p}{\partial w_i} = \frac{w_i(\Sigma w)_i}{\sigma_p}.$$

Since

$$\sum_{i=1}^{N} RC_i = \sigma_p,$$

the equal-risk-contribution condition is

$$RC_i = \frac{\sigma_p}{N}, \qquad i = 1, \ldots, N. \tag{8}$$

Equivalently,

$$w_i(\Sigma w)_i = w_j(\Sigma w)_j \qquad \text{for all } i, j. \tag{9}$$

In the general case of correlated returns, risk parity does not have a simple closed-form solution and is usually computed numerically by solving

$$\min_w \sum_{i=1}^{N} \sum_{j=1}^{N} [w_i(\Sigma w)_i - w_j(\Sigma w)_j]^2$$

subject to

$$\mathbf{1}' w = 1, \qquad w_i \geq 0.$$

However, if Σ is diagonal such that assets are uncorrelated and

$$\Sigma = \text{diag}(\sigma_1^2, \ldots, \sigma_N^2),$$

then

$$RC_i = \frac{w_i^2 \sigma_i^2}{\sigma_p}.$$

Equalizing risk contributions implies

$$w_i^2 \sigma_i^2 = w_j^2 \sigma_j^2 \qquad \text{for all } i, j,$$

which yields

$$w_i \sigma_i = w_j \sigma_j.$$

Thus

$$w_i^{RP} \propto \frac{1}{\sigma_i}. \tag{10}$$

After normalization,

$$w_i^{RP} = \frac{\sigma_i^{-1}}{\sum_{j=1}^{N} \sigma_j^{-1}}. \tag{11}$$

I use the inverse-volatility portfolio as a simple approximation to risk parity.

G Baseline Portfolio Grid Results

Table 18: Baseline Portfolio Performance Across Correlation Grid

Metric	Equal Weight	Market Composition	Risk Parity
Expected Return (%), mean	4.798	4.846	4.815
Volatility (%), min	9.327	9.222	9.286
Volatility (%), median	9.933	9.808	9.886
Volatility (%), max	10.505	10.360	10.451
Sharpe Ratio, min	0.124	0.130	0.126
Sharpe Ratio, median	0.131	0.137	0.133
Sharpe Ratio, max	0.139	0.146	0.142
Real Return (%), mean	2.798	2.846	2.815
Coefficient of Variation, min	1.944	1.903	1.929
Coefficient of Variation, median	2.070	2.024	2.053
Coefficient of Variation, max	2.189	2.138	2.170
Left Tail Proxy (%), mean	5.658	5.593	5.635
Upside Proxy (%), mean	6.094	6.013	6.065
Skew Proxy (%), mean	0.436	0.419	0.430
Implied Risk Aversion γ, min	-0.104	-0.086	-0.095
Implied Risk Aversion γ, median	0.041	0.067	0.049
Implied Risk Aversion γ, max	0.213	0.266	0.232

The inference procedure relies on least squares with a small sample (at most 5 assets), thus the negative implied risk aversion reported for the minimums across candidate portfolios and in the subsequent table reflects the frailty of the procedure more than genuine risk seeking.

H Correlation Grid Portfolio Analytics

This appendix reports the full portfolio comparison across the five asset correlation grid[22]. See the main text for median weights.

Table 19: Five Asset Correlation Grid Portfolio Performance

	60/40 Portfolios					Global / Baseline Portfolios				
Metric	EW	GMV	MVO	Max ER	RP	Baseline	EW	GMV	MVO	RP
Expected Return (%), mean	4.920	5.010	5.540	5.540	4.973	4.846	4.920	4.933	5.900	4.900
Volatility (%), min	11.115	9.058	10.903	10.903	10.530	9.766	11.115	4.451	18.200	6.279
Volatility (%), median	11.290	9.385	11.199	11.199	10.725	9.766	11.290	4.739	18.200	6.625
Volatility (%), max	11.462	9.661	11.488	11.488	10.916	9.766	11.462	5.105	18.200	6.953
Sharpe, min	0.124	0.155	0.178	0.178	0.135	0.138	0.124	0.257	0.132	0.201
Sharpe, median	0.126	0.161	0.182	0.182	0.137	0.138	0.126	0.307	0.132	0.211
Sharpe, max	0.128	0.170	0.187	0.187	0.140	0.138	0.128	0.331	0.132	0.223
Real Return (%), mean	2.920	3.010	3.540	3.540	2.973	2.846	2.920	2.933	3.900	2.900
Coeff. Var., min	2.259	1.799	1.968	1.968	2.117	2.015	2.259	0.902	3.085	1.281
Coeff. Var., median	2.295	1.873	2.022	2.022	2.157	2.015	2.295	0.950	3.085	1.352
Coeff. Var., max	2.330	1.924	2.074	2.074	2.195	2.015	2.330	1.060	3.085	1.419
Left Tail Proxy (%), mean	5.800	5.712	5.240	5.240	5.736	5.593	5.800	2.423	7.400	3.811
Upside Proxy (%), mean	6.300	6.078	5.600	5.600	6.202	6.013	6.300	2.489	8.000	4.072
Skew Proxy (%), mean	0.500	0.366	0.360	0.360	0.466	0.419	0.500	0.066	0.600	0.261
Implied γ, min	-0.025	0.025	0.014	0.014	-0.018	0.001	-0.025	-25.486	0.010	-0.092
Implied γ, median	-0.023	0.026	0.015	0.015	-0.016	0.004	-0.023	-14.045	0.011	-0.075
Implied γ, max	-0.022	0.028	0.017	0.017	-0.014	0.008	-0.022	-6.677	0.012	-0.062

[22]For several portfolios, the reverse-optimized γ is negative or does not match the value passed into the MVO routine. This is not a contradiction. The input γ governs the MVO portfolios only. When I back out γ from non-MVO portfolios, I effectively identify the value of γ that best rationalizes that portfolio as if it came from a mean-variance model. Even for MVO portfolios, fitted γ need not match the input exactly. The calibration procedure includes an intercept term, whereas the optimization routine imposes constraints directly.

I New Portfolio Return–Volatility Plots

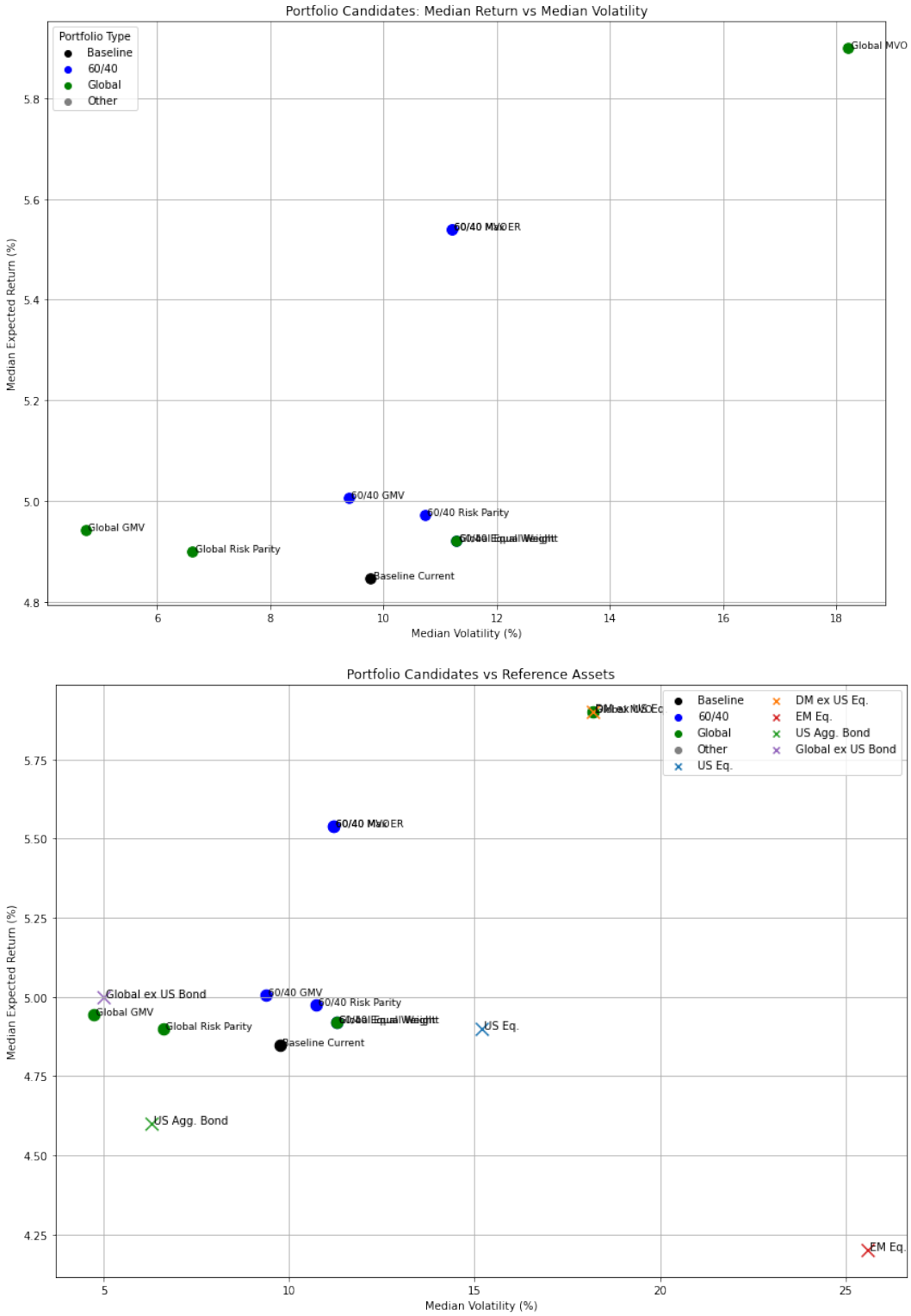

Figure 3: Portfolio and reference asset return vs. volatility.